Energy-based neural constitutive surrogates for the nonlinear hyperelastic RVE

Anisotropy, local tangent scaling, and a constructive polyconvex PANN

July 27, 2026

Abstract

This memorandum documents a controlled constitutive-learning study for the two-dimensional nonlinear hyperelastic representative volume element (RVE). Three models are considered: a flexible energy PANN in material components of C, the same type of PANN in locally isotropised tangent coordinates, and a polyconvex PANN. The last is a new constructive model: it is objective, stress-free in the reference configuration, anisotropic without imposing D_4, energy–stress consistent, polyconvex in $(F, \operatorname{cof} F, \det F)$, and has a volumetric barrier. All stresses are obtained by differentiation of the same scalar energy. The three models are trained from Stage–1 trajectories only and evaluated on Stage–10.

1 Purpose and modelling decision

The goal is not to learn three stress-time curves separately. The target is one macroscopic stored-energy density W of the RVE. Once W is known, the second Piola stress follows by differentiation. This makes the energy and the stress mutually consistent by construction.

The RVE is treated here as a *general anisotropic* material. In particular, no square symmetry is assumed. A D_4 model would require the additional identity

$$W(C_{11}, C_{22}, C_{12}) = W(C_{22}, C_{11}, -C_{12}), \tag{1}$$

which is not used below. This is a modelling decision, not a statement that the square-shaped numerical cell makes the effective material isotropic.

2 Kinematics, stress convention, and objectivity

Let

$$C = F^T F, \qquad E = \frac{1}{2}(C - I), \qquad J = \det F = \sqrt{\det C} > 0. \tag{2}$$

The database and all neural models use the engineering-shear coordinate

$$\boldsymbol{e} = \begin{bmatrix} E_{11} \\ E_{22} \\ \gamma_{12} \end{bmatrix}, \qquad \gamma_{12} = 2E_{12}. \tag{3}$$

The work identity is therefore

$$dW = S_{11}\, dE_{11} + S_{22}\, dE_{22} + S_{12}\, d\gamma_{12}, \qquad \frac{\partial W}{\partial \boldsymbol{e}} = \begin{bmatrix} S_{11} \\ S_{22} \\ S_{12} \end{bmatrix}. \tag{4}$$

Equation (4) is crucial: the derivative returns precisely the second Piola components stored in the direct FOM data; there is no factor-two ambiguity in the shear component and no independently trained stress network.

Objectivity concerns a rotation of the *current* spatial observer. For every proper orthogonal Q,

$$F \mapsto QF \quad \Longrightarrow \quad (QF)^T(QF) = C. \tag{5}$$

Consequently, a scalar model built from C (or from the objective minor norms used in Section 5) satisfies $W(QF) = W(F)$. This is different from material symmetry: material symmetry would rotate the RVE relative to its fixed material axes and is not imposed.

3 Anisotropic representation

Let a_0 and b_0 be two physical directions attached to the reference RVE. The independent material contractions in two dimensions are

$$c_{aa} = a_0 \cdot C a_0, \qquad c_{bb} = b_0 \cdot C b_0, \qquad c_{ab} = a_0 \cdot C b_0. \tag{6}$$

In the stored RVE coordinates they are C_{11}, C_{22}, and C_{12}. There are only three independent components of a symmetric 2×2 tensor; "six invariants" is the corresponding count in three dimensions. We also pass J explicitly, although it is redundant through $J^2 = c_{aa}c_{bb} - c_{ab}^2$. It separates the volumetric state visibly from the other features.

Under a relabelling of reference axes, C, a_0, and b_0 must be transformed together. The three numbers in (6) are then unchanged. This is coordinate covariance, not an extra material symmetry.

4 Two flexible energy PANNs

4.1 C-PANN

The accuracy-oriented reference model is a smooth Softplus MLP $\widehat{W}_\theta$ of

$$\boldsymbol{z}_C = [c_{aa} - 1,\ c_{bb} - 1,\ c_{ab},\ J - 1]^T. \tag{7}$$

It is made reference-consistent using the affine correction

$$W_C(\boldsymbol{e}) = \widehat{W}_\theta(\boldsymbol{z}_C(\boldsymbol{e})) - \widehat{W}_\theta(\boldsymbol{z}_C(\boldsymbol{0})) - \left.\frac{\partial \widehat{W}_\theta}{\partial \boldsymbol{e}}\right|_{\boldsymbol{0}} \cdot \boldsymbol{e}. \tag{8}$$

Thus $W_C(\boldsymbol{0}) = 0$ and $\partial W_C/\partial \boldsymbol{e}|_{\boldsymbol{0}} = \boldsymbol{0}$, up to floating-point roundoff. This correction is legitimate for a flexible PANN, but it would generally destroy a polyconvexity certificate. Hence this model makes *no* convexity or polyconvexity claim.

4.2 PANN-T: local anisotropic-to-isotropic tangent scaling

The second flexible model tests the idea in the anisotropic mixed-formulation paper of Rossi, Zorrilla, and Codina [4]. At the reference state, the FOM finite-difference tangent is

$$D_0 = \left.\frac{\partial \boldsymbol{S}}{\partial \boldsymbol{e}}\right|_{\boldsymbol{0}}. \tag{9}$$

In the coordinates (3), the closest isotropic tangent that preserves the uniform-dilation modulus is

$$\widehat{D} = \begin{bmatrix} \lambda + 2\mu & \lambda & 0 \\ \lambda & \lambda + 2\mu & 0 \\ 0 & 0 & \mu \end{bmatrix}, \quad \kappa = \frac{D_{00} + 2D_{01} + D_{11}}{4}, \quad \mu = \frac{D_{00} - 2D_{01} + D_{11} + D_{22}}{5}, \quad \lambda = \kappa - \mu. \tag{10}$$

For positive definite D_0 and $\widehat{D}$, define

$$T = \widehat{D}^{-1/2} D_0^{1/2}, \qquad D_0 = T^T \widehat{D} T. \tag{11}$$

PANN-T uses $[(T\boldsymbol{e})^T/\varepsilon_{\mathrm{scale}}, J - 1]^T$ as the MLP input and applies the same reference correction as (8). Numerically, $\|D_0 - \widehat{D}\|_F / \|D_0\|_F = 0.222\%$ and $\|T - I\|_F / \|I\|_F = 0.374\%$. Thus it is an intentionally small, local conditioning transformation for this RVE. It does not make the nonlinear material isotropic, and it does not imply convexity or polyconvexity.

5 Constructive anisotropic polyconvex PANN

Polyconvexity is a condition on the deformation gradient:

$$W(F) = G(F, \operatorname{cof} F, J), \qquad G \text{ convex in } (F, \operatorname{cof} F, J). \tag{12}$$

It is *not* synonymous with convexity in E or with a positive definite $\partial \boldsymbol{S}/\partial \boldsymbol{e}$.

For each $k = 1, 2, 3$, choose three unit structural directions d_{ki} and positive weights w_{ki} satisfying

$$\sum_{i=1}^{3} w_{ki}\, d_{ki} \otimes d_{ki} = I. \tag{13}$$

The three direction systems are

$$(0^\circ, 50^\circ, 130^\circ), \qquad (20^\circ, 83^\circ, 151^\circ), \qquad (12^\circ, 76^\circ, 143^\circ),$$

with the positive weights recorded in the reproducibility file. They have no 90°-rotation closure and no D_4 average is taken. From them we form

$$\begin{aligned} I_1 &= \operatorname{tr} C = |F|^2, \\ Q^F_{k,p} &= \sum_{i=1}^{3} w_{ki}\,(d_{ki} \cdot C d_{ki})^p = \sum_{i=1}^{3} w_{ki} |F d_{ki}|^{2p}, \qquad p = 2, 3, \\ Q^H_{k,p} &= \sum_{i=1}^{3} w_{ki}\,(d_{ki} \cdot \operatorname{cof}(C) d_{ki})^p = \sum_{i=1}^{3} w_{ki} |\operatorname{cof}(F) d_{ki}|^{2p}, \qquad p = 2, 3. \end{aligned} \tag{14}$$

The evaluated features are

$$\boldsymbol{z}_{\mathrm{pc}} = (I_1, \{Q^F_{k,2}, Q^F_{k,3}, Q^H_{k,2}, Q^H_{k,3}\}_{k=1}^3, J, J^2). \tag{15}$$

Let Φ_θ be an input-convex neural network (ICNN) whose input and hidden-to-hidden weights are non-negative and whose activations are Softplus. It is then convex and coordinate-wise non-decreasing in $\boldsymbol{z}_{\mathrm{pc}}$. The energy is

$$W_{\mathrm{pc}}(F) = \Phi_\theta(\boldsymbol{z}_{\mathrm{pc}}) - \Phi_\theta(\boldsymbol{z}_{\mathrm{pc}}(I)) - r \log J + \frac{\beta}{2}(J-1)^2, \qquad r, \beta > 0. \tag{16}$$

The sixth-power terms are not cosmetic. In two dimensions, a balanced quartic aggregate by itself obeys an accidental 90° identity: if $t_i = d_i \cdot C d_i$, then the rotated values are $\operatorname{tr}(C) - t_i$ and the balance relation makes $\sum_i w_i(\operatorname{tr}(C) - t_i)^2 = \sum_i w_i t_i^2$. The cubic aggregate $\sum_i w_i t_i^3$ does not obey this identity in general. It therefore supplies anisotropic information while its first derivative at $C = I$ remains isotropic.

Why this is polyconvex. For a fixed d, $|Fd|^2$ is convex in F and the maps $x \mapsto x^p$ for $p = 2, 3$ are convex and increasing for $x \geq 0$; hence $|Fd|^{2p}$ is convex in F. Exactly the same argument holds for $|\operatorname{cof}(F)d|^{2p}$ as a function of the cofactor minor. Positive sums, I_1, J, and J^2 are convex functions of the corresponding minor variables on $J > 0$. The monotone convex-composition rule therefore makes the ICNN term convex in $(F, \operatorname{cof} F, J)$. Finally, $-r \log J$ and $\beta(J-1)^2/2$ are convex in $J > 0$. This proves (12); it is a construction proof, not a numerical penalty in the loss.

Why the reference stress vanishes without breaking the proof. At $C = I$, relation (13) makes the first derivative of each $Q^F_{k,p}$ and $Q^H_{k,p}$ isotropic. The same is true of I_1, J, and J^2. Thus the ICNN has a reference derivative of the form $(p, p, 0)$ in the coordinate (3). We set

$$r = p = \frac{1}{2}\left(\frac{\partial \Phi_\theta}{\partial E_{11}} + \frac{\partial \Phi_\theta}{\partial E_{22}}\right)\Bigg|_{E=0}. \tag{17}$$

Since $\partial \log J / \partial \boldsymbol{e}|_0 = (1, 1, 0)$, the logarithmic term cancels this pressure. Equation (16) consequently gives $W_{\mathrm{pc}}(I) = 0$ and $\boldsymbol{S}(I) = \boldsymbol{0}$, while retaining polyconvexity. Moreover, it diverges as $J \to 0^+$ and has positive quadratic growth as $J \to \infty$.

6 Training protocol and what is not imposed

All models minimize a combined relative squared-error loss on direct FOM energy and direct FOM second Piola stress:

$$\mathcal{L} = 0.65\, \mathcal{L}_W + 0.35\, \mathcal{L}_S + 0.65\, \mathcal{L}_{S,\mathrm{comp}}, \tag{18}$$

where $\mathcal{L}_W$ and $\mathcal{L}_S$ normalize the mean squared energy and all-component stress errors by their respective mean squared targets, and $\mathcal{L}_{S,\mathrm{comp}}$ is the equally weighted mean of the three componentwise normalized stress errors. In every term the predicted stress is the derivative in (4) of the predicted energy.

All three final models use all 109 460 saved states of Stage–1 trajectories 1–10. The certified ICNN is trained in CPU float64 because its energy–stress loss differentiates the energy twice. No training script opens Stage–10.

We deliberately do not impose

$$\frac{\partial \boldsymbol{S}}{\partial \boldsymbol{e}} \succ 0. \tag{19}$$

The independent FOM tangent audit gives a minimum eigenvalue of -0.0648 GPa on a Stage–10 state. Thus a global version of (19) would contradict the FOM response. Polyconvexity is the relevant stronger-in-F construction for the third model, but it does not require a positive strain Hessian.

7 Held-out Stage–10 assessment

Table 1 reports relative L^2 errors against the direct FOM labels. The direct HPROM–ANN row is a compatible reconstruction of energy and the same second Piola stress at its stored displacement state; it is not a separately retrained comparison.

Table 1: Stage–10 relative L^2 errors against direct FOM labels.

Model	W	S	S_{11}	S_{22}	S_{12}
Flexible C-PANN	2.077%	4.551%	4.062%	5.001%	9.477%
PANN-T (local tangent metric)	0.061%	0.114%	0.130%	0.095%	0.902%
Certified polyconvex PANN	1.848%	2.643%	2.064%	3.125%	9.375%
Direct HPROM–ANN reconstruction	0.000%	0.118%	0.109%	0.117%	13.338%

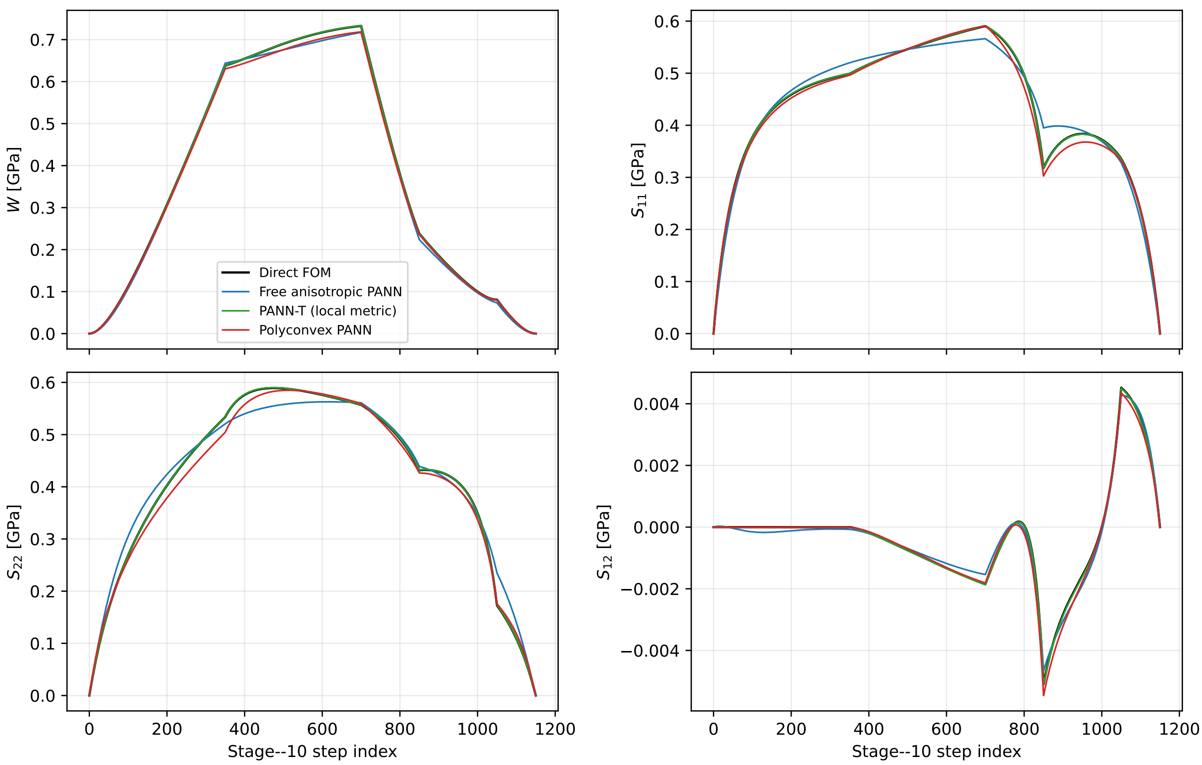

Figure 1: Held-out energy and energy-conjugate second Piola components along Stage–10. Every PANN stress shown here is differentiated from the displayed PANN energy.

The evaluator additionally verifies the reference conditions, objectivity under $F \mapsto QF$, the absence of an imposed D_4 average, and (for the certified model) numerical rank-one curvature samples. The smallest sampled rank-one second derivative is $4.609e-01$ GPa. This sample is not the proof—the proof is the construction above—but it is a useful attempt to falsify an implementation error. The learned volumetric coefficients are $r = 3.0728e-01$ and $\beta = 3.1221e-05$ in normalized energy units. A separate broad positive-definite sampling audit found a minimum energy of $5.888e-04\,\mathrm{GPa}$; this is a useful physical check, not a proof of global non-negativity.

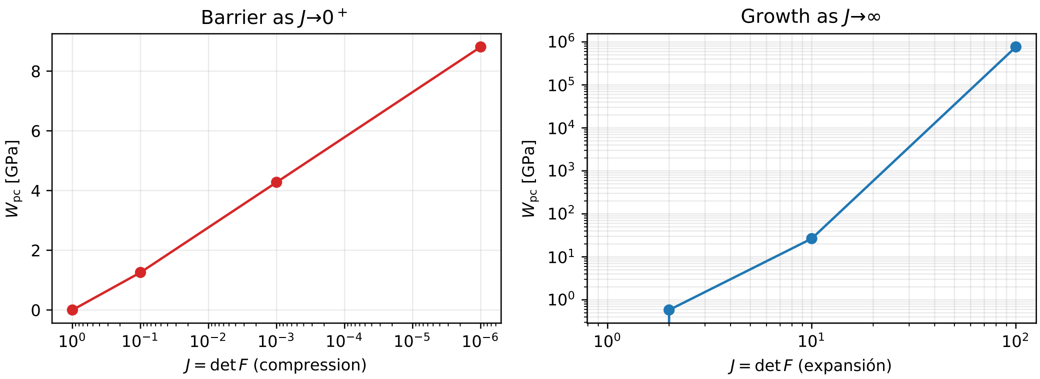

Figure 2: Volumetric response of the certified PANN: the logarithmic barrier acts as $J \to 0^+$ and the positive quadratic term supplies growth at large J.

8 Conclusions and recommended use

The three models answer three distinct questions and should not be presented as interchangeable black boxes. The C-PANN is the flexible accuracy baseline. PANN-T tests a principled local tangent scaling; it changes the conditioning, not the physical symmetry class or the global mathematics. The third model is the mathematically strongest candidate: its objectivity, reference state, energy–stress consistency, anisotropic structural response, polyconvexity, and volumetric barrier are architectural properties. Its prediction gap, if present in Table 1, is the honest cost of those restrictions rather than a guarantee silently omitted from the model.

References

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